

DXRD Help

The **DXRD** program suite is for accurately computing **X-ray diffraction from perfect single crystals** using the **X-ray dynamical theory**. It consists of the following programs.

1. Bragg-case Darwin Curve of Single Crystal Diffraction (for plane wave)

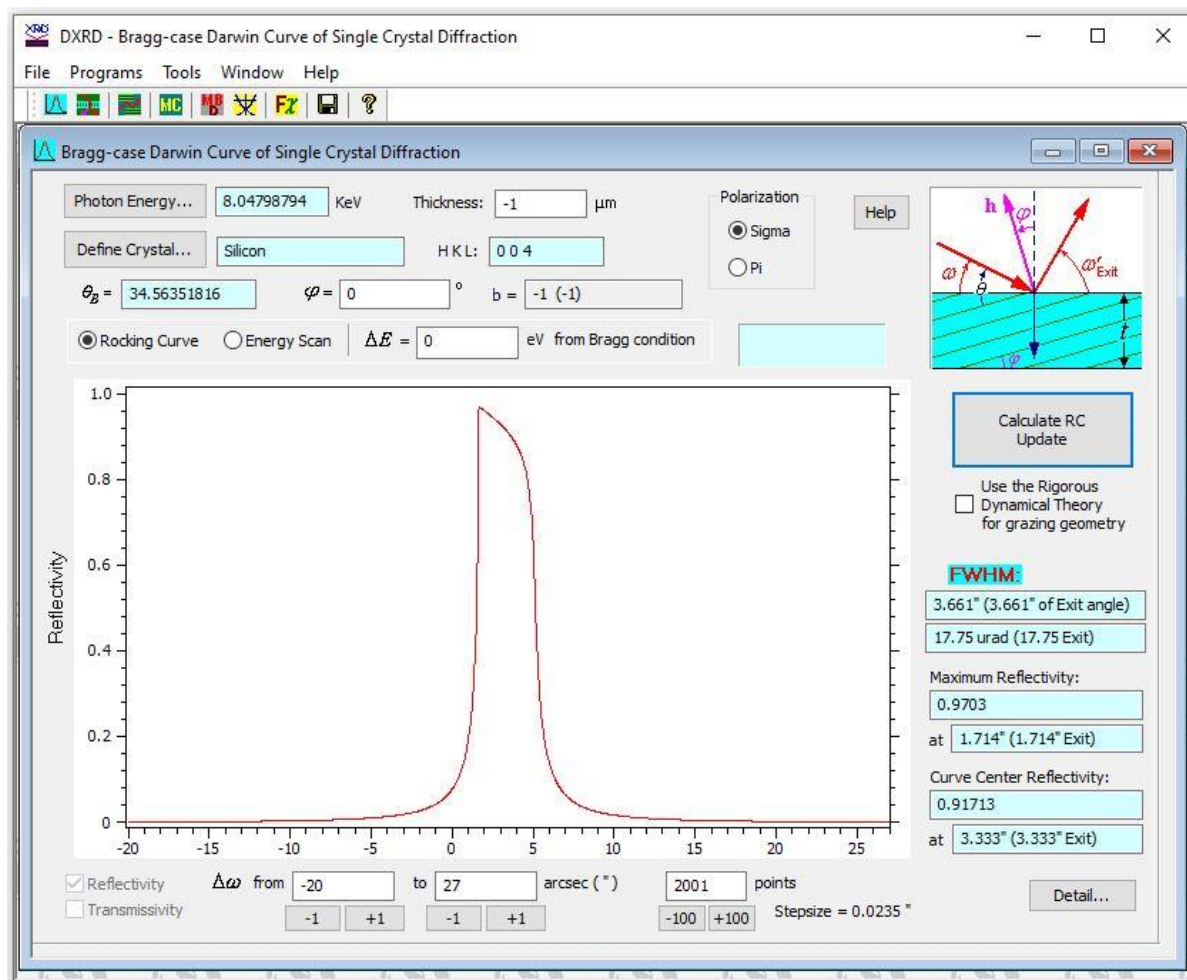


Figure A1. The Bragg-case Diffraction program.

This program allows you to calculate the **Darwin Curve** of **Bragg-case X-ray diffraction** from a semi-infinite-thickness (-1) crystal or a thin crystal plate. The Curve can be a function of either the **Incidence Angle (Rocking Curve)** or the **Photon Energy (Energy Scan, showing the Bandpass/Bandwidth of a monochromator)**.

Click the "**Photon Energy**" button to set the photon energy (**must be < 100 keV**) by selecting from a list of characteristic X-ray energies, or **by directly typing any energy value you want**. The "**Define Crystal**" button allows you to import a **Crystal Structure** and then select the **Reflection Indices HKL** you want. **DXRD** does not require registering crystal structures to a central database. Instead, it allows you to compile a simple text file ***.struct** for any crystal structure. The file only requires the information of (1) a crystal name, (2) the lattice constants $a, b, c, \alpha, \beta, \gamma$, (3) the number of atoms in the unit cell, (4) the positions of the atoms in the unit cell, and (5) the Debye characteristic temperatures. You can use **Notepad** or any text editor to open the structure file of silicon, **Si.struct**,

in the main directory to see the format of the structure file. In the sub-directory "**\CrystStructFiles**" you can find many structure files for various crystals. You can easily define your own structure files based on the format. As an alternative, you can also use the **Tools -> "Editor of Crystal Structure"** to define crystal structure files.

The program then calculates the **Structure Factors** and the corresponding susceptibility components [called the **Chi (χ) Factors**]. By default, **DXRD** uses **tabulated coefficients** to calculate structure factors for photon energies < 100 keV, which is similar to the XOP package (<https://www.aps.anl.gov/Science/Scientific-Software/XOP>, <https://www.esrf.fr/Instrumentation/software/data-analysis/xop2.4>). DXRD also provides another program to "**Calculate Structure Factors of Crystals Using Brennan's Method**" [S. Brennan and P. L. Cowan, *Rev. Sci. Instrum.* **63**, 850 (1992), <https://doi.org/10.1063/1.1142625>]. This method may give slightly different structure factors for **Sigma (σ) and Pi (π) polarization** states. Therefore, the "**Define or Modify Crystal and its Parameters**" dialog has additional **π -polarization χ factors**. The tabulated-coefficient method gives the same χ factors for both σ - and π -polarization. You can also **manually set your own χ factors** from other sources in the "**Define or Modify Crystal and its Parameters**" dialog.

When you calculate the rocking curve, the energy is set to the Photon Energy (**Bragg energy**) you have set. However, the ΔE parameter allows you to slightly deviate the energy from the Bragg energy, so that you can see how the rocking curve shifts with energy. **The incidence angle is denoted by ω and the exit angle is ω'_{exit} in Figure A1. ω'_{exit} may simply be written as ω' .** The **horizontal axis $\Delta\omega$** is always relative to the **Bragg Angle θ_B** calculated with the **Photon Energy without ΔE** . For **Energy Scan**, the **Incidence Angle** is fixed at θ_B by default, but you can slightly deviate it from θ_B by setting the $\Delta\omega$ edit box. The horizontal axis for Energy Scan is **ΔE** relative to the **Photon Energy**.

If the photon energy is set to be lower than the back-reflection energy of the reflection (corresponding to $\theta_B = 90^\circ$, the lowest energy for activate the Bragg reflection), the Bragg angle θ_B is set to 90° and may be shown as **90 (O+). Note that when θ_B approaches 90° , the Darwin curve generally becomes very wide and strongly depends on small photon energy changes.**

The **Asymmetric Angle φ** is **positive** along the direction in the diagram, and negative along the other direction. For nearly backward diffraction, determining the sign of φ may be difficult, but just make sure that the incidence angle is $\omega_B = \theta_B - \varphi$ and the exit angle is $\omega_B' = \theta_B + \varphi$.

The Asymmetric Factor is defined here as

$$b = - \frac{\sin \omega_B'}{\sin \omega_B}$$

Note that in convention, the asymmetric factor was defined as

$$b = - \frac{\sin \omega_B}{\sin \omega_B'}$$

So we list both values in the "**b =**" edit box.

When you change any of the parameters, you need to click on the "**Calculate RC Update**" button to update the results. Afterwards, **you may click on any point on the Curve, and there is a box at the top-right of the curve showing the coordinates of the clicked point.**

If the crystal has a **finite thickness ($t > 0$)**, you can also plot the **Transmissivity Curve** by checking the "**Transmissivity**" checkbox at the bottom-left corner.

For **grazing-incidence/exit geometry** where the Incidence or Exit angle is very small, close to the critical angle of X-ray specular reflection, the conventional X-ray dynamical theory could fail. So, the program provides a check box “**Use The Rigorous Dynamical Theory**” under the “**Calculate RC Update**” button to calculate the Curve using the so-called “**Extended Dynamical Theory**”. For non-grazing geometry, this method gives the same result (checking/unchecking the checkbox does not matter), but for grazing geometry, the results may be noticeably different.

The box below the sign “**FWHM**” shows the FWHM width of the calculated Curve. Note that for **Asymmetric Reflection Geometry** ($b \neq -1$), the incidence and exit angle have different divergence. So, the box **first shows the FWHM of the rocking curve in terms of the incidence angle, followed by the FWHM in terms of the exit angle (indicated by “Exit”) in the parentheses**. The unit here is **arcsec**. For Energy Scan, the FWHM is the bandwidth, but the value in the parentheses is still the angular FWHM in terms of the exit angle. The reason is that, for a fixed incidence angle, the exit angle still varies with the photon energy for asymmetric reflection --- **dispersion!** (No dispersion for symmetric Bragg reflection geometry.) The next edit box shows the same results but in the unit of **μrad**.

A complete Curve has the **Maximum Reflectivity peak**, besides which there are two **Half-Maximum** points. The midpoint of these two points is defined as the **Curve Center** and its reflectivity is the **Curve Center Reflectivity**.

You can use Menu “**File → Export RC Data**” to save the Curve data to a file. If the crystal has a **finite thickness** ($t > 0$), you can also export the **Transmissibility**. If you have checked “**Use the Rigorous Dynamical Theory**”, you can also export the **Specular Reflectivity** (usually very small except for grazing geometry).

Clicking the “**Detail...**” button gives you more information about the Curve, though most of the time you may not be interested in those data.

The “**Search Reflections**” button let you search for all the Bragg reflections of the current crystal that (1) satisfy the chosen **Bragg angle (Theta)** within the energy range (“**E from**”, “**to**”) keV, or (2) satisfy the chosen **Energy (E)** within the angular range (“**Theta from**”, “**to**”) degrees. For cubic Si/Ge/C structures, the (up to eight) reflections ($\pm h, \pm k, \pm l$) are equivalent reflections. When you check “**Exclude Equivalents for Si/Ge/C**”, only one of reflections, (h, k, l), is presented, which is for the purpose of making the list short and concise. When the crystal Name is “**Si**”, “**Silicon**”, “**Ge**”, “**Germanium**”, “**C**”, or “**Diamond**”, this button is automatically checked, but you can uncheck it. For any other crystal name, this button is automatically unchecked, but you can manually check it if you are sure the current structure is the type of cubic diamond structure. If this button is unchecked, no eligible reflection is excluded (so that all equivalents will be shown, which may make the list very long). The “**Bandwidth Cal. Range**” is the range (and step) the program uses to calculate the **energy-dependent reflectivity curve** of each reflection, with the zero point being the Bragg energy E_B (i.e., the geometric Bragg energy without the correction of the small refraction effect). Most of the time you do not need to change these parameters, but you can change them if necessary. The (angular) Darwin Width $\Delta\theta$ is calculated from the bandwidth ΔE by $\Delta\theta = \tan \theta_B (\Delta E / E_B)$, **which is accurate for $\theta_B < 90^\circ$** . For $\theta_B \rightarrow 90^\circ$ (back reflection), the angular Darwin curve could be very wide and complicated, and this equation may fail. But you can use the current “Bragg-case Darwin Curve of Single Crystal Diffraction” program to accurately calculate the angular Darwin curve. This program applies to any two-beam coplanar Bragg case diffraction.

2. Laue-case Single Crystal Diffraction (for Plane Wave)

This program is similar to the above Bragg-case program. The major difference is that the “**Incident Angle ω** ” and the “**Exit angle ω'** ” are defined differently with respect to the Crystal Surface, see Figure A2. Also see the definition of the **Asymmetric Angle φ** . The crystal thickness for the Laue case must always be finite (> 0 , NOT -1). The Reflectivity (R) Curve for the Laue cases show oscillation fringes in most cases. In particular, thick crystals have rapid reflectivity oscillation. For a constant incidence angle, the reflectivity also changes periodically with crystal thickness, which is called the “**Pendellösung**” phenomenon. The periodicity is called the “**Pendellösung distance**”, denoted by $\tau(\Delta\omega)$. When the

crystal thickness is $t = N\tau$, the central reflectivity becomes 0. The central reflectivity is maximized when at $t = (N + \frac{1}{2})\tau$. Here $N = 0, 1, 2, \dots$ is an integer. In the “**Detail...**” dialog, both the analytical “**Pendellösung distance**” and “**Half Pendellösung distance**” derived from analytical equations are provided at the Curve Center. The analytical values may be accurate for low-absorption cases. Numerical calculations of τ has not been implemented yet.

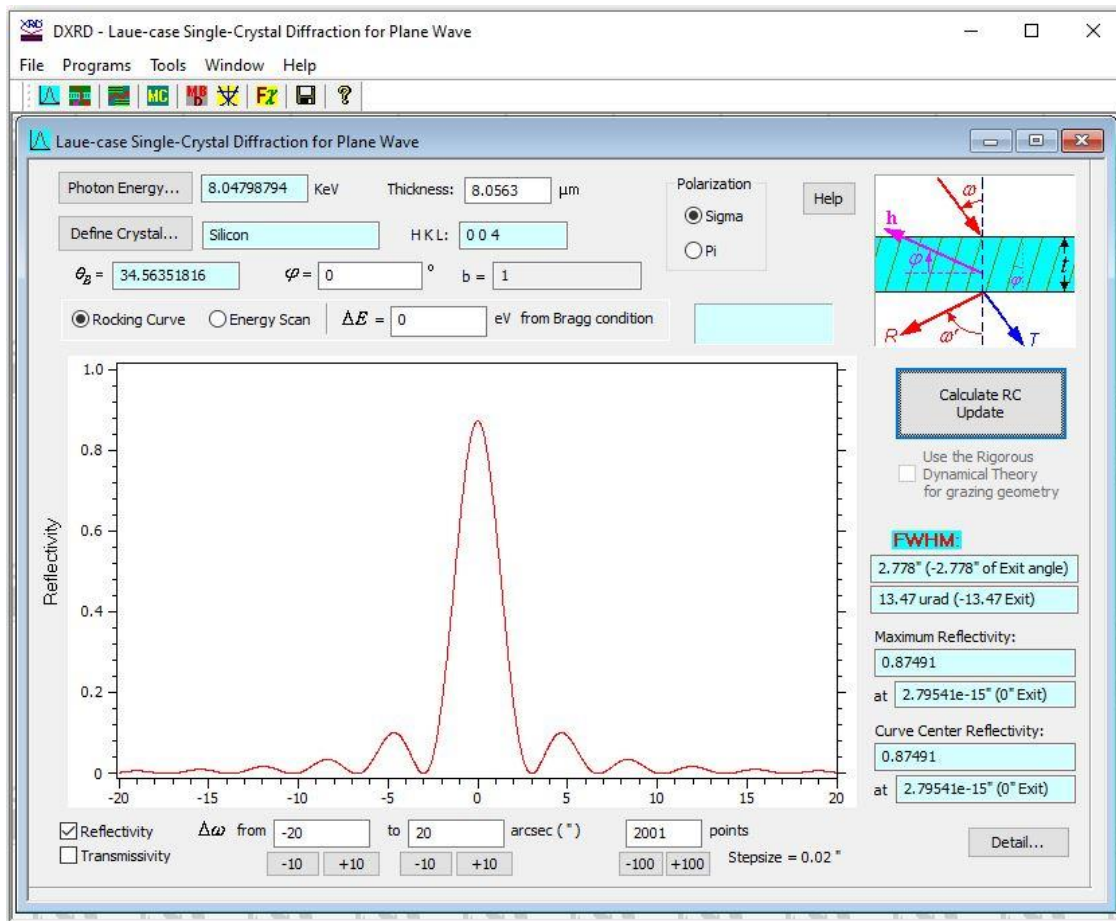


Figure A2. The Laue-case Diffraction program.

3. Convoluted Rocking Curve for Divergent & Polychromatic Incident Beam

The above two programs are for the **plane-wave incident beam**. In practical experiments, the incident beam always has **divergence** and energy spread (**spectral bandwidth**), which can **make the measured rocking curve MUCH WIDER than the intrinsic Darwin curve** in most cases. This program is for calculating the crystal rocking curve based on an X-ray beam with finite divergence and energy spread.

The default set-up of this program is that the divergent and polychromatic incident beam is first monochromatized by a fixed **double-crystal monochromator (DCM)**, and then incident on a sample. **The rocking curve is measured (computed) with the DCM fixed while you rock the Sample.** This is the typical setup at synchrotron beamlines. The incident spectrum is assumed to be the Gaussian spectrum with a spectral bandwidth of the **FWHM** value that you can set. Usually, the X-ray beam upstream the DCM is a white beam, so you can give set a very large FWHM value. The bandpass of a DCM is usually 1 eV, so as long as the FWHM is large enough, its detailed values have little effect on the results. The **Spectrum Range** is the Energy Range you use for integration over the energy. You can choose a large Spectrum Range value with adequate

points. The **divergence of the incident beam** may greatly affect the final rocking curve width. Therefore, you have to carefully set this value based on your beamline parameters (including the slits used).

Before a full computation, you must make sure that the parameters you set are adequately large enough to achieve convergence of the integration and meanwhile, the computing fast enough. To do so, you can keep increasing each parameter, and compare the results with the previous results. The program can **plot both the current rocking curve and the previous curve** by **checking “Show Previous RC”**. **When there is no noticeable difference between the current and previous curves, it is likely that you reach the convergence requirements [for the testing parameter(s)]**. But in general, the **Beam Divergence** parameter is a real condition of the beam and you should not change it significantly, though you can freely change the **Points**.

For the non-dispersive configuration, when the Monochromator and the Sample are of the same kind of crystal and have the same Bragg reflection (or the two Bragg angles are very close to each other), the convoluted rocking curve width is nearly independent of the divergence and spectrum of the incident beam. This is usually called “double-crystal rocking curve”. It is a very accurate and reliable method for measuring the rocking curve widths of crystals. This method is accurate even if the incident beam is a laboratory source with large divergence. **The disadvantage of this method is that for different samples/reflections, you have to switch to different monochromators (if they are available)**, which is very inconvenient. For a Lab-source-based double-crystal diffractometer, the DCM may not be necessary, a single monochromator may be enough. In this case, you can **uncheck the “DCM” choice**. For Cu K α 1 source, we suggest the following typical parameters: **Random polarization** (half Sigma and half Pi), **spectral FWHM 2.6 eV**, the **Spectrum Range 15 eV**, and the **Beam Divergence** about ~ 1.5 mrad.

As is well known and can be easily demonstrated by the above **“Bragg-case Darwin Curve of Single-Crystal Diffraction”** program, the diffraction peak is always **slightly shifted upward** (typically by a few arcsecs) from the geometrical Bragg angle θ_B due to the small X-ray **refraction** effect. Therefore, the **principal incident angle of the monochromator** is not exactly the Bragg angle. It should be corrected by the small Bragg peak shift, and the **Angular Center** parameter must be equal to this shift. You can use the **“Bragg-case Darwin Curve of Single-Crystal Diffraction”** program to quickly find the Bragg peak shift (i.e. the **relative Curve Center position**) of the monochromator and fill the **Angular Center** with the shift value.

You can also choose the **“Dispersive Configuration”** if necessary. For a DCM, the **Non-dispersive or Dispersive Configuration** is based on the **relationship between the second crystal of the DCM and the Sample**.

You can check the **“Channel Cut”** checkbox below the **“Define Sample...”** button if the sample crystal is a channel-cut crystal (which is not common though).

A very simple and useful method to check the condition of a synchrotron DCM is to rock the 2nd crystal to measure the rocking curve width. This is a standard “double-crystal rocking curve measurement”. For example, in the diagram of Fig. 3. You can set the Photon Energy to 10 keV, select the first crystal as Si 111 and **de-select the DCM checkbox. Then select the 2nd crystal (**Sample**) also as Si 111. Using the default parameters, you can calculate the FWHM of the rocking curve width as 38.5 μ rad. This is very close to typical measurements. Note that this value is slightly wider than the intrinsic Si 111 Darwin width of 29 μ rad calculated by the above “Bragg-case Darwin Curve of Single Crystal Diffraction (for plane wave)” program.**

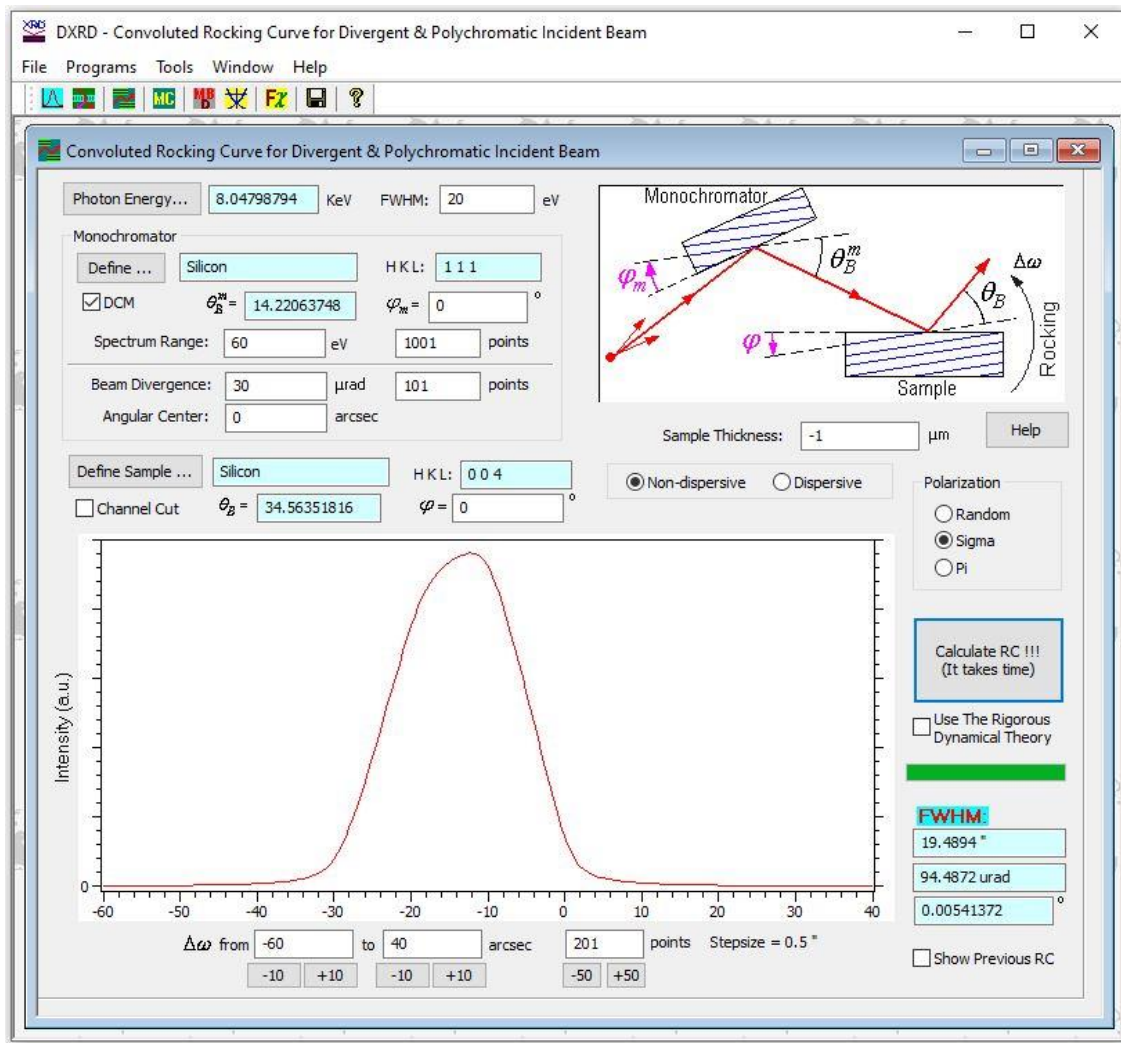


Figure A3. Convolved Rocking Curve for Divergent & Polychromatic Incident Beam program. Note that here the DCM is of typical Si 111 reflection. Si 004 reflection has a much broader Rocking curve width (19.5 arcsecs) than that in Figure A1. However, if we change the DCM reflection to the same reflection of the sample, Si 004, then the rocking curve width of the sample becomes 4.37 arcsecs (not shown), very close to that in Figure A1.

4. Multiple-Crystal Monochromator

The goal of this program is to calculate the **bandwidth** and **angular acceptance** of **arbitrary multiple-crystal monochromators**, but the development is still in the early stage.

First, if you check the “**Only 1st**” checkbox, you can compute the rocking curve or bandwidth of the **1st crystal** just like the “**Bragg-case Darwin Curve of Single Crystal Diffraction**” program does. But here you can also check the “**Channel-cut**” checkbox of the **first crystal** to compute the rocking curve of a channel-cut crystal.

If you uncheck “**Only 1st**”, you can bring in the **2nd crystal**, which again can be either a single crystal or a channel-cut. You can set the second crystal in the “**Non-dispersive**” or “**Dispersive**” setup. Although the schematic diagram is similar to that of the above “**Double-Crystal Diffraction**” setup, there is a major difference here. That is, here the relative orientations between the two crystals are fixed, and the incident beam is a plane wave. You can only scan the **Incidence Angle** or Incident **Photon Energy**, similar to the program “**Bragg-case Darwin Curve of Single Crystal Diffraction**”. By

comparison, in the “**Double-Crystal Diffraction**” setup, the incident beam is not a plane wave. It is a divergent beam with a finite spectral bandwidth. Meanwhile, the 1st crystal is fixed and **you can only rotate the 2nd crystal**.

As mentioned before, the diffraction peak of a Bragg-case reflection **shifts slightly toward the high-angle direction** from the **geometrical Bragg angle** due to the slight X-ray refraction effect. To compute multiple-crystal diffraction, you must “align” each of the crystal to the peak position by taking into account these slight peak shifts. The **$\Phi = \text{angle}$** is the relative orientation between the two crystals calculated without taking into account the slight peak shift; it is only based on the two geometrical Bragg angles and the asymmetric cut angles. The **$\Delta\Phi = \text{angle}$** is the overall corrections calculated from the slight peak shifts of the two crystals. You can click the “**Auto Set...**” button to let the program calculate the angles. Then you may manually **tweak** it to optimize the overall reflectivity or bandwidth. Or you can input the values yourself (from single-crystal Darwin curve calculations). **Note that if you do not set the two angles correctly, the reflectivity (efficiency) of the monochromator may be very low!**

After all the parameters of the crystals and their relative orientations are set, you can perform the **Angular Scan** or **Energy scan** of the **entire monochromator**, similar to the “**Bragg-case Darwin Curve of Single Crystal Diffraction**” program. For Energy Scan, the “ **$\Delta\omega =$** ” editbox allows you to slightly deviate the incidence angle (of the first crystal) from the Bragg angle, mainly for compensating the slight diffraction peak shift. **Setting it to 0** usually does not affect the Curve shape or width, but may induce a slight shift of the entire Curve.

A typical application of this program is that you set both crystals as channel cuts and place them in the **dispersive** configuration. Then it becomes a **Bartels Monochromator**, and you can calculate its **bandwidth** and **angular acceptance**. Especially you can vary the **asymmetric angles** (of both crystals) to tune the bandwidth and angular acceptance --- Don’t forget to set and tweak the **$\Delta\Phi =$** angle correctly, otherwise the efficiency would be very low.

The menu “Tools -> Parasitic Reflections of Back reflections” is to find the parasitic reflections of any back-reflection ($\theta_B \rightarrow 90^\circ$) of only cubic structures, see the reference for details. The CMBDPs (Continuous Multiple-Beam Diffraction Planes) and HIDs [Horizontal Incidence Directions, i.e., the direction of the projection of the incident beam on the primary (HKL) planes] are for you to avoid Multiple-Beam Diffraction by avoiding these directions. Note that only 111 and 220 reflections of cubic diamond have no parasitic reflections.

5. Dynamical-Theory Computation of Multiple-Beam Diffraction

6. Multiple-Beam Diffraction Lines (Monochromator Glitches) in Azimuth-Energy Map

The **Help File** of these two programs is “**MBD-Help.pdf**”. You can open it directly in the main folder of DXRD, or you can open the “**Dynamical-Theory Computation of Multiple-Beam Diffraction**” program and click the “**Help**” button.